Heart Disease Prediction Model

Using Machine Learning Classification Algorithms

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AI/ML Internship Project Report

1. Introduction

In the present healthcare systems, cardiovascular diseases remain one of the leading causes of mortality worldwide. The World Health Organization (WHO) reports that an estimated 17.9 million people die each year from cardiovascular diseases, accounting for approximately 31% of all global deaths. Early detection and prediction of heart disease can significantly reduce these numbers by allowing timely medical intervention.

Predicting heart disease risk in advance can help patients as well as their healthcare providers to take preventive measures and necessary steps for treatment. Many healthcare institutions have adopted data-driven diagnostic systems today. Such systems are favorable to the medical professionals in identifying high-risk patients early. The purpose of building a machine learning-based prediction system is to assist doctors and health workers in screening patients who may be at higher risk of heart disease based on their clinical parameters.

The core function of the Heart Disease Prediction system is to help the patient to know his/her heart disease risk in advance by using multiple Machine Learning classification algorithms. Such techniques would help patients to take proactive measures for their health and would enable doctors to identify those individuals who might need immediate medical attention and further diagnostic tests.

Machine Learning classification algorithms are widely used in medical diagnosis applications. This is because machine learning models can identify complex patterns in clinical data that may not be immediately apparent to human observers. The models used in this project include Logistic Regression, Random Forest, Support Vector Machine (SVM), K-Nearest Neighbors (KNN), Decision Tree, and Gradient Boosting Classifier.

2. Problem Statement

The Main Objective of the "Heart Disease Prediction Application" is to implement a robust machine learning model that predicts whether a patient is at risk of heart disease based on clinical and diagnostic parameters collected during medical examinations.

The "target" variable (1 = heart disease present, 0 = no heart disease) is our label (output) and the rest of the clinical features serve as our input features. The dataset contains 13 input features: age, sex, chest pain type (cp), resting blood pressure (trestbps), serum cholesterol (chol), fasting blood sugar (fbs), resting electrocardiographic results (restecg), maximum heart rate achieved (thalach), exercise induced angina (exang), ST depression induced by exercise (oldpeak), slope of the peak exercise ST segment (slope), number of major vessels colored by fluoroscopy (ca), and thalassemia results (thal).

The problem statement can be defined as follows: "Given a dataset containing clinical attributes of patients, using the features available from the UCI Heart Disease dataset, define and evaluate multiple classification algorithms to identify whether a patient is at risk of heart disease in the near future, and also to evaluate different machine learning models on the dataset to determine the best-performing algorithm." The dataset contains 303 patient records with 14 attributes collected from the Cleveland Clinic Foundation.

The project also involves building an interactive web application using Streamlit that allows users (patients or doctors) to input clinical parameters and receive real-time heart disease risk predictions.

3. Methodology

1. Data Collection: The UCI Heart Disease Dataset (Cleveland database) containing 303 patient records with 14 clinical attributes was used.

2. Data Preprocessing: Missing value handling, feature scaling using StandardScaler, and train-test split (80:20 ratio) with stratified sampling.

3. Exploratory Data Analysis (EDA): Target variable distribution, age and gender-based analysis, correlation heatmap, feature distributions, and categorical feature analysis.

4. Model Training: Six ML algorithms were trained - Logistic Regression (C=1.0), Random Forest (n_estimators=100), SVM (kernel=RBF), KNN (n_neighbors=7), Decision Tree (max_depth=8), and Gradient Boosting (n_estimators=100).

5. Model Evaluation: Accuracy, Precision, Recall, F1-Score, ROC-AUC Score, 5-Fold Cross-Validation, and Confusion Matrix analysis.

6. Web Application Development: An interactive Streamlit web application was built for real-time predictions.

4. Results and Discussion

4.1 Exploratory Data Analysis

The dataset analysis revealed that the target variable is fairly balanced, with approximately 54.5% of patients having heart disease (165 cases) and 45.5% being healthy (138 cases). This balanced distribution is favorable for model training as it prevents class imbalance issues.

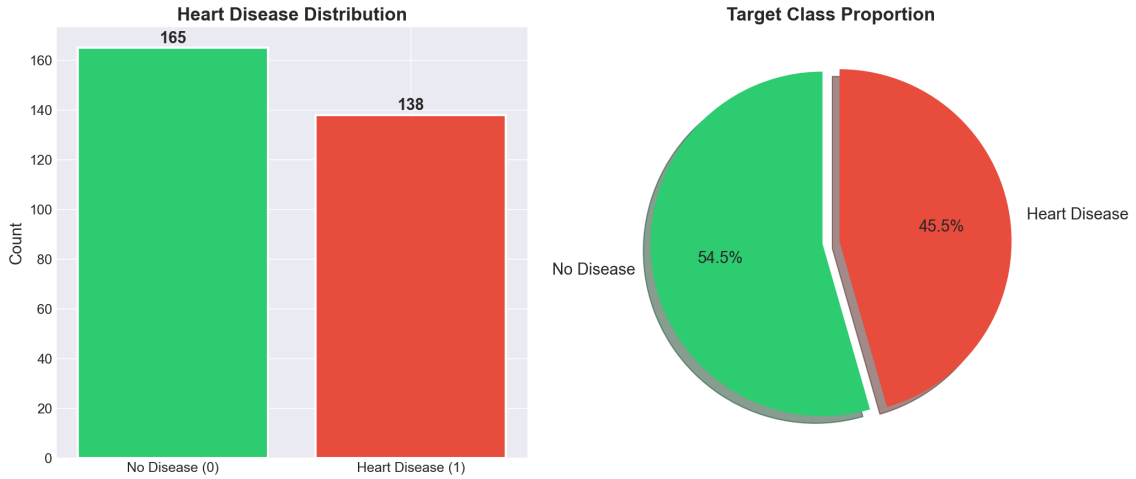

Figure 1: Target Variable Distribution - Heart Disease vs No Disease

4.2 Age Distribution Analysis

The age distribution analysis shows that heart disease patients are spread across a wide age range, with peak occurrences in the 50-60 age group. The overlapping distributions indicate that age alone is not a definitive predictor.

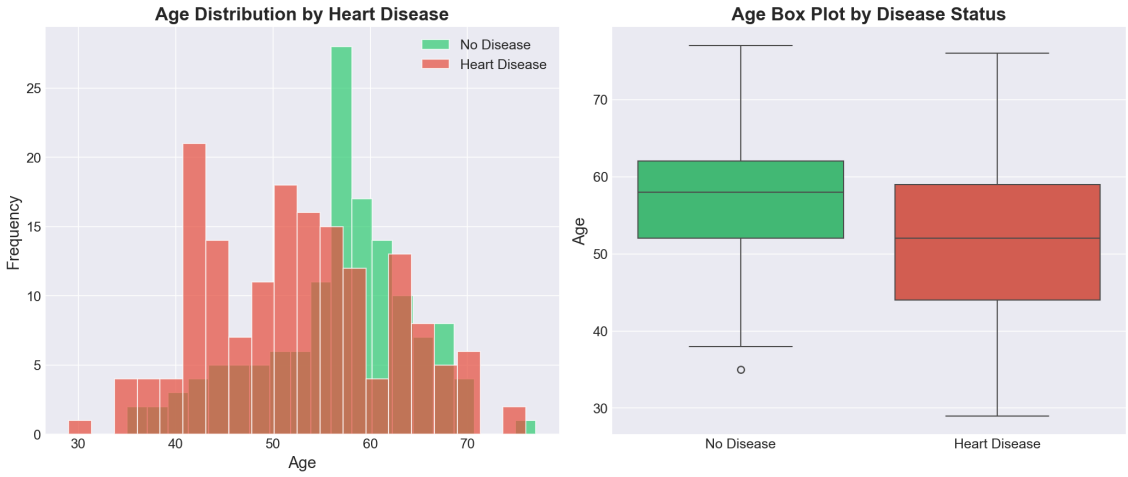

Figure 2: Age Distribution by Disease Status

4.3 Gender Analysis

Males show a higher prevalence of heart disease in the dataset compared to females. The dataset contains more male patients (207) than female patients (96). Gender-based analysis reveals distinct patterns in disease occurrence.

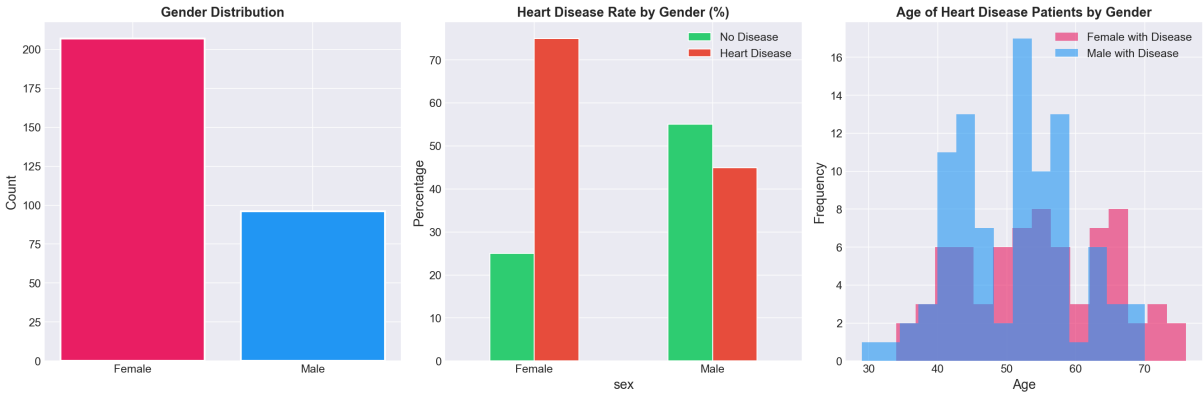

Figure 3: Gender-based Heart Disease Analysis

4.4 Feature Correlation Analysis

The correlation heatmap reveals the relationships between all features. Features like "cp" (chest pain type), "thalach" (max heart rate), "exang" (exercise angina), and "oldpeak" (ST depression) have the strongest correlations with the target variable.

Figure 4: Feature Correlation Matrix Heatmap

4.5 Feature Distributions

The distribution of each feature across the dataset shows varying patterns. Understanding these distributions helps in feature engineering and model selection.

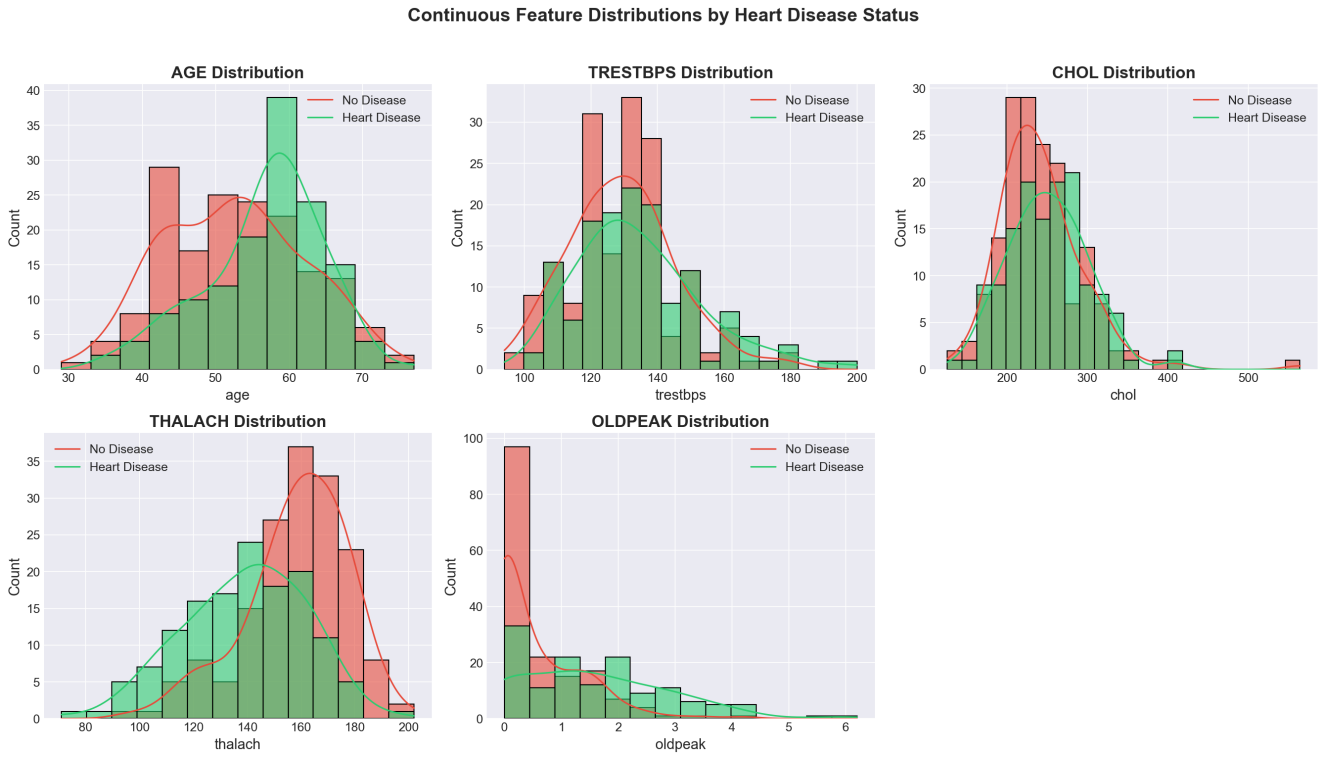

Figure 5: Distribution of All Features

4.6 Categorical Feature Analysis

Categorical features such as chest pain type, fasting blood sugar, resting ECG, exercise-induced angina, slope, and thalassemia show distinct patterns when analyzed against the target variable.

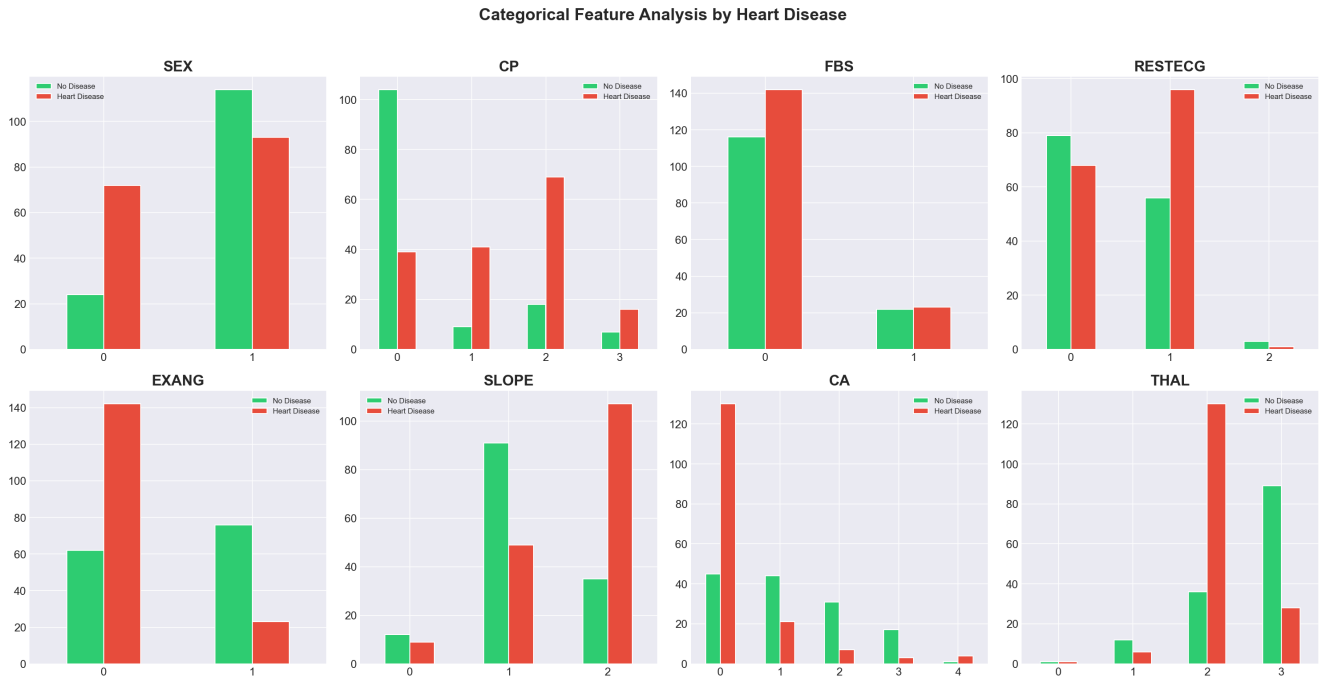

Figure 6: Categorical Features Analysis by Disease Status

4.7 Model Performance Comparison

All six models were trained and evaluated on the test set (20% of data). The results demonstrate that Random Forest achieved the highest accuracy of 83.6%, followed by SVM and KNN at 82.0%. Logistic Regression achieved 80.3% accuracy. The ensemble methods and linear models performed comparably well on this dataset.

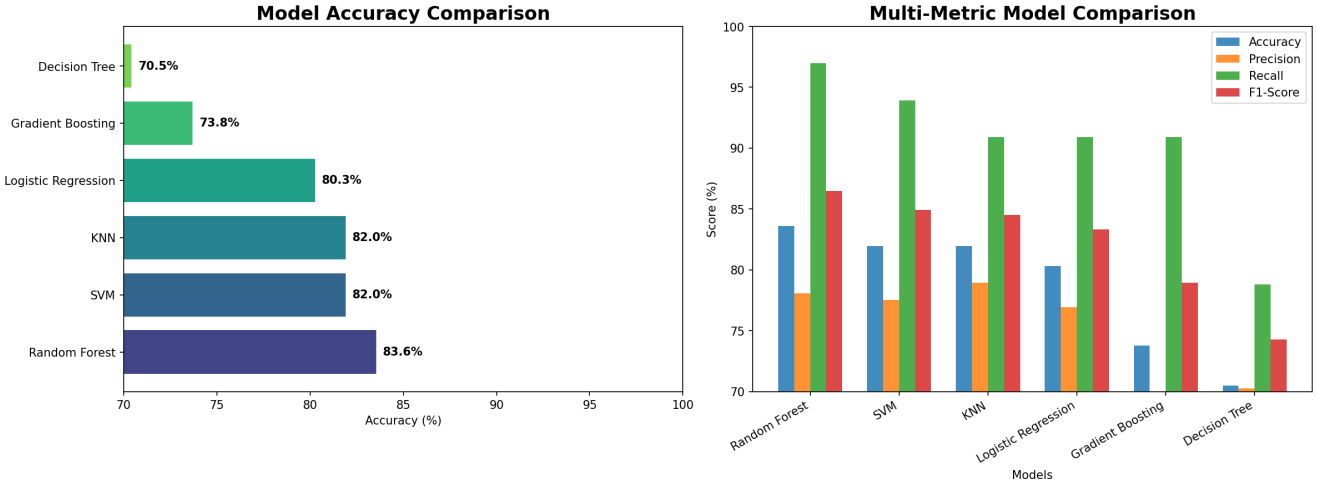

Figure 7: Model Accuracy and Multi-Metric Comparison

4.8 Confusion Matrix Analysis

The confusion matrices show the true positive, true negative, false positive, and false negative predictions for each model. Random Forest shows the best balance between precision and recall.

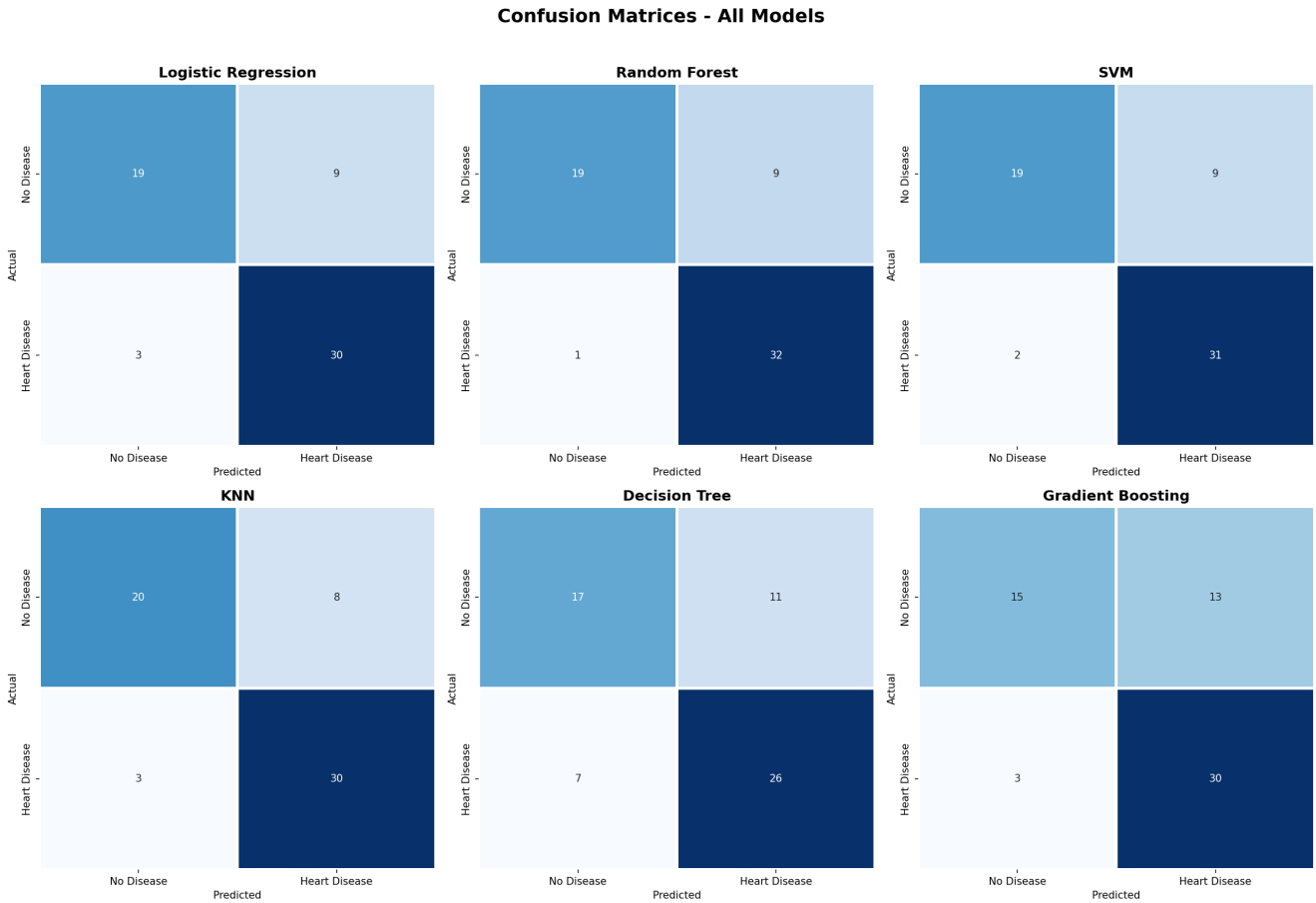

Figure 8: Confusion Matrices for All Models

4.9 ROC Curve Analysis

The ROC curves show the trade-off between true positive rate and false positive rate for each model. Higher AUC scores indicate better model performance. Random Forest achieved the highest AUC of 0.909.

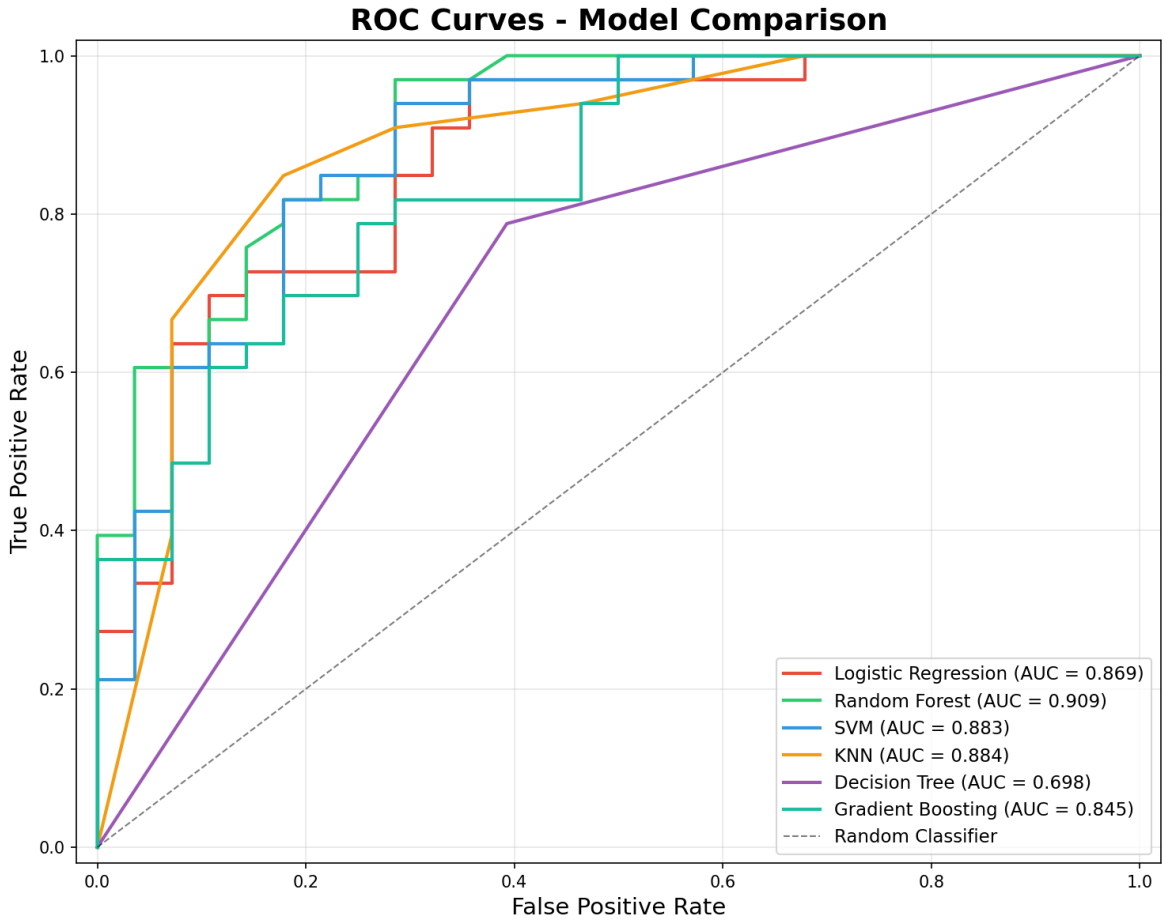

Figure 9: ROC Curves - Model Comparison

4.10 Feature Importance Analysis

Feature importance from Random Forest and Gradient Boosting reveals that the most important features for prediction are: thalach (Max Heart Rate), cp (Chest Pain Type), ca (Number of Major Vessels), oldpeak (ST Depression), thal (Thalassemia), and age. These findings align with medical literature.

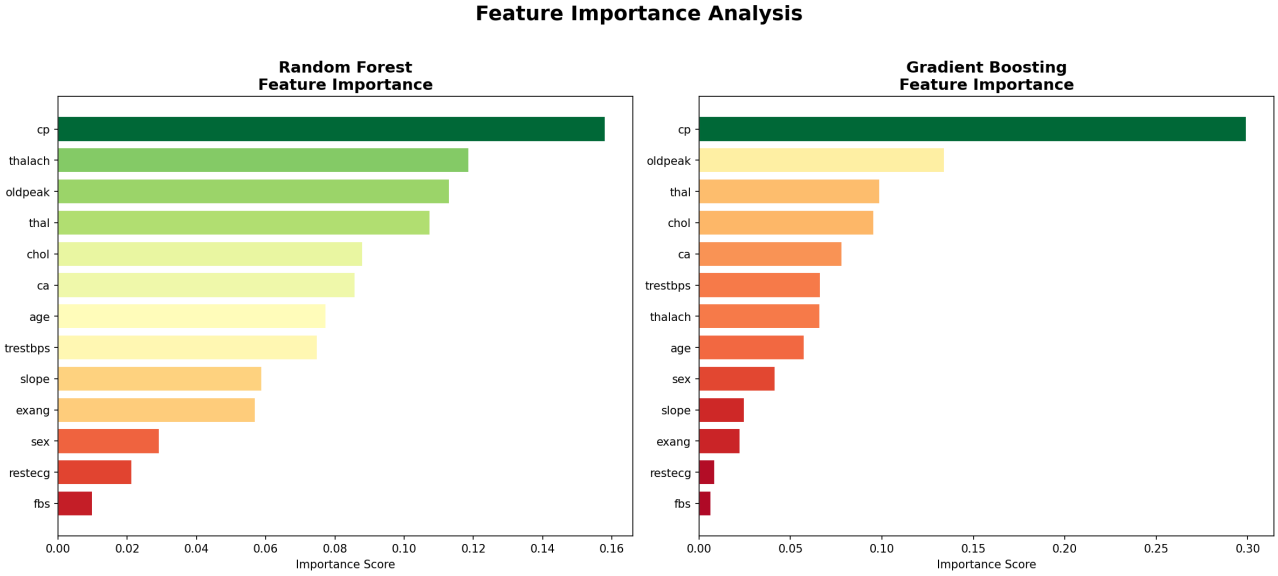

Figure 10: Feature Importance - Random Forest & Gradient Boosting

4.11 Cross-Validation Scores

Five-fold cross-validation scores confirm that the models generalize well and are not overfitting. The standard deviation of CV scores was low across all models, indicating stable and reliable performance.

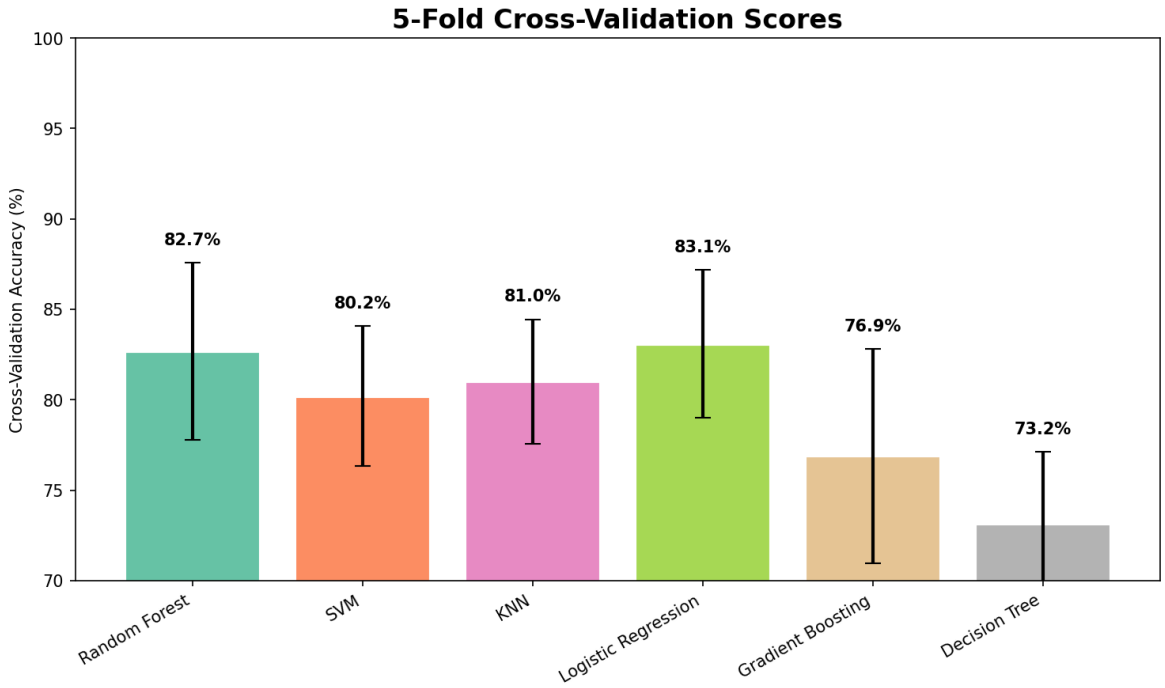

Figure 11: 5-Fold Cross-Validation Accuracy Scores

5. Conclusion

In this project, we performed a comprehensive analysis of heart disease prediction using multiple machine learning algorithms on the UCI Cleveland Heart Disease dataset. The project involved complete data preprocessing, extensive exploratory data analysis with rich visualizations, training of six different classification models, thorough evaluation using multiple metrics, and deployment through an interactive Streamlit web application.

The analysis revealed several clinically significant patterns in the data, including the strong predictive power of maximum heart rate, chest pain type, and number of major vessels colored by fluoroscopy. The correlation analysis and feature importance rankings provided insights that align well with established medical knowledge about cardiovascular disease risk factors.

After evaluating all the algorithms on different parameters including accuracy, precision, recall, F1-score, ROC-AUC, and cross-validation scores, the best-performing model was Random Forest with 83.6% accuracy and 0.909 ROC-AUC score. Logistic Regression and ensemble methods consistently showed strong performance.

The interactive Streamlit web application allows end-users (patients or healthcare professionals) to input clinical parameters and receive instant heart disease risk predictions with probability scores and visual risk indicators. This makes the model accessible and practically useful for preliminary screening purposes.

It is important to note that this model is intended as a screening tool and should not replace professional medical diagnosis. In the future, the system can be enhanced by incorporating larger datasets, adding deep learning models, implementing patient history tracking, deploying on cloud platforms, and adding API endpoints for hospital integration.

6. References

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